

Abstract

The “CapScan” project is a desktop-based software developed in Python to verify the authenticity, capacity, and reliability of memory cards and USB drives. It provides users with an easy and efficient way to analyze storage devices and detect counterfeit, low-quality, or corrupted drives that may misrepresent their actual specifications. The system offers a simple, user-friendly interface designed to guide users through various testing processes with clarity and accuracy. Key features of CapScan include verifying real storage capacity, performing read and write speed tests, checking data integrity, and generating detailed reports for better understanding of device performance. It also allows users to store and retrieve previous test results, making it easier to compare device performance over time and track potential issues. The system provides customizable viewing options for test outputs, enabling clearer presentation and interpretation of results. The advantages of CapScan include improved reliability when selecting storage devices, enhanced protection against data loss and corruption, and strengthened confidence in the authenticity of memory cards and USB drives. It reduces the risk of system errors caused by faulty devices, ensures that users are informed before trusting a storage medium with important data, and ultimately supports safer, more efficient digital data management.

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Chapter: 1 | INTRODUCTION

1.1. Problem Specification

In our modern lives, a memory card or a simple USB drive is often the vault for our most cherished memories and critical work—from wedding photos and final academic papers to crucial business reports. We entrust these tiny devices with everything. However, a major, hidden threat is undermining this trust: the widespread proliferation of counterfeit and deliberately deceptive storage devices.

Imagine losing years of irreplaceable data because the brand-new 128GB flash drive you bought for a bargain was actually a tiny 8GB drive programmed to lie. This is the reality facing countless users today. As one expert report noted, "Fake storage devices... are a scourge and have probably lost countless people's data. They're usually a smaller-capacity device that's been reprogrammed to trick the computer into thinking it's bigger than it actually is. When you try to save data that's too big for the real capacity, old data gets overwritten so your files get corrupted." [1]

This deception leads to catastrophic data loss, file corruption, and frustrating system failures. Currently, the average user lacks the simple, reliable tools needed to spot these digital imposters before disaster strikes. CapScan is designed to restore that trust. It steps in as an essential, user-friendly guardian, providing dependable capacity checks, performance verification, and integrity validation to ensure that the hardware you buy is the hardware you can rely on.

1.2. Objectives

The core mission of CapScan is to establish a secure and dependable method for verifying the authenticity and reliability of storage devices, ensuring that users can place complete confidence in their hardware before storing sensitive personal, academic, or professional data. By integrating comprehensive testing mechanisms and a user-friendly interface, CapScan aims to fundamentally eliminate the risks associated with counterfeit, misreported, or failing storage devices. The system is meticulously designed to empower users with unimpeachable accuracy, clear performance insights, and meaningful diagnostics that support safe and informed decision-making. The enhanced objectives of CapScan include:

- **Counterfeit Detection:** Validating True Capacity. To accurately validate a device's true usable capacity, using thorough write-and-read tests to immediately expose storage devices that falsely advertise higher capacities. This objective guarantees users receive genuine, trustworthy hardware and are proactively protected from fraudulent or misrepresented products.

- **Performance Clarity:** Benchmarking Real-World Speeds. To measure and benchmark data transfer speeds (read and write) under conditions that simulate real-world usage. This provides users with a transparent understanding of the device's overall efficiency, enabling the clear identification of slow, low-quality, or degraded storage media.
- **Integrity Assurance:** Protecting Stored Data. To perform rigorous read/write operations across the entire accessible space of the device. This process reliably detects latent corruption or unstable sectors, which is crucial for safeguarding the accuracy and permanence of stored data and proactively preventing unexpected data loss.
- **Device Health Monitoring:** Predictive Failure Analysis. To leverage S.M.A.R.T. (Self-Monitoring, Analysis, and Reporting Technology) data (where applicable for larger devices) to offer an advanced health assessment. This enables the early and predictive detection of potential hardware failures by monitoring critical parameters such as temperature, reallocated sector counts, and uncorrectable read error rates.
- **Accessible Reporting:** Driving Informed Decisions. To provide detailed, organized, and professionally formatted reports that succinctly summarize the findings from capacity verification, performance results, data integrity checks, and device health status. This delivers users the definitive clarity required to make well-informed hardware management decisions.
- **User-First Design:** Delivering Seamless Accessibility. To deliver an intuitive, clean, and robust interface-based platform. This ensures effortless navigation and immediate accessibility for all user demographics, ranging from beginners seeking simple reassurance to IT professionals requiring precise, actionable diagnostic information.

1.3. Scope

The purpose of CapScan is to serve as a practical, accessible, and dependable tool for testing everyday storage devices. It is designed to give users confidence in the hardware they rely on, whether for personal files, academic work, or professional data.

Our Focus: The User's Digital Assurance

CapScan is a desktop-based Python application built for seamless use on standard computer systems. Its primary focus is the analysis of the most commonly used portable storage devices—USB flash drives and SD cards. The system centers on delivering three essential guarantees:

- **Capacity Verification:** Ensuring that the storage device genuinely provides the capacity it claims to offer.
- **Performance Benchmarking:** Measuring real-world read and write speeds to reveal the device's true performance level.

- **Data Reliability Check:** Evaluating the stability and health of the memory chips to ensure that stored data remains accurate and secure.

CapScan is designed for a broad range of users—from students safeguarding important academic files to IT professionals evaluating multiple devices for deployment. Anyone who needs to confirm the authenticity, reliability, or safety of storage hardware can depend on CapScan as an essential diagnostic tool.

Our Limitations: Where We Draw the Line

To ensure that CapScan remains fast, accessible, and easy to operate, its functions are intentionally limited to software-based diagnostics. For this reason, the scope of CapScan does not include:

- **Advanced Data Recovery:** CapScan identifies issues but does not retrieve lost or deleted data.
- **Firmware Repair:** The system cannot modify or reprogram the internal controller of a storage device.
- **Specialized Hardware Diagnostics:** CapScan does not perform proprietary or hardware-level testing that requires specialized equipment.

By remaining strictly focused on software-based testing, CapScan provides a practical and highly accessible solution that requires no additional tools or technical expertise. It is designed to be the user's first, most trustworthy line of defense against counterfeit, faulty, or unreliable storage devices.

1.4. Organization Of Project Report

The various chapters that make up this report's structure each focus on important project aspects:

- **Chapter 1** lays the groundwork by introducing the problem, defining objectives, and outlining the scope of CapScan.
- **Chapter 2** provides the theoretical foundation and reviews existing literature related to storage device verification, ensuring a solid understanding of the concepts that support the system.
- **Chapter 3** analyzes the system requirements, presenting both functional and non-functional needs, along with a feasibility study to confirm CapScan's practical viability.
- **Chapter 4** details the system design, including architectural diagrams, interface layouts, and component interactions that guide the application's development.
- **Chapter 5** explains the implementation process and describes the testing procedures used to validate CapScan's performance, accuracy, and reliability.
- **Chapter 6** presents the results, offers conclusions based on the project's achievements, and proposes future improvements and enhancements.

Chapter: 2 | BACKGROUND

2.1 Existing System Analysis

The verification of USB drives and memory cards is traditionally performed using basic system features or outdated third-party tools. While these methods offer partial insight, they fail to deliver complete, accurate, and reliable authentication of storage devices. The following subsections highlight the limitations of widely used existing solutions.

2.1.1. Manual Verification Using User Observation

A large number of users rely on simple, informal checking techniques. This includes inserting the storage device, reviewing the displayed capacity, copying a few files, or trusting the device based on packaging, brand stickers, or physical build quality.

Pros:

- No additional tools or installation needed.
- Very quick and convenient for basic checks.
- Accessible to all users without technical skills.
- Useful for identifying only very obvious device failures.

Cons:

- Cannot detect spoofed or “expanded” fake capacities created via firmware manipulation.
- File copying only tests a tiny portion of the memory, leaving most of the storage unverified.
- No mechanism to detect bad sectors or hidden memory corruption.
- Does not measure performance or speed drops caused by low-quality chips.
- Users easily trust external branding, which counterfeit products often replicate perfectly.
- Fails completely for advanced storage fraud where the device pretends to have more space than actually present.

2.1.2. Generic Operating System Tools

Built-in utilities such as Windows File Explorer, Disk Management, macOS Disk Utility, and Linux `fdisk`, `lsblk`, `dmesg`, or basic SMART readers offer limited diagnostic features.

Pros:

- Pre-installed with every operating system.
- Can format, view partitions, or detect file system errors.
- Provides simple access to device properties.
- Useful for handling common disk issues like minor corruption or formatting failures.

Cons:

- OS tools always trust the reported capacity, even when the flash controller lies.
- Cannot inspect the internal structure of flash memory or its physical limits.
- No secure method to test full write/read integrity.
- Cannot identify counterfeit USB controllers or reprogrammed chips.
- Does not provide performance benchmarking under controlled conditions.
- Only detects superficial filesystem problems, not hardware defects or fraudulent behavior.

2.1.3. Software That Claims “Impossible Repairs”

Many free tools online claim to “fix 0MB USB drives,” “restore original size,” “repair fake flash,” or “recover hidden storage.” These tools often rely on risky low-level operations such as forcing firmware resets, rewriting controller tables, or modifying partition maps.

Pros:

- Sometimes helpful for minor, non-physical firmware issues.
- Appeals to users who want a quick and simple repair.
- May temporarily restore access to corrupted partitions.

Cons:

- Cannot repair physically defective or counterfeit memory chips.
- Often destroy existing data due to unsafe formatting methods.
- Incorrectly lead users to believe fake devices can be “repaired” into real ones.
- No accurate capacity verification or performance measurement.
- Many of these tools are outdated, unsafe, unstable, or developed by unknown sources.
- Some contain malware or aggressive advertising.
- Mislead users into reusing unreliable storage devices, increasing risk of future data loss.

2.1.4. Manufacturer-Specific Diagnostic Tools

Storage manufacturers provide official diagnostic software, such as:

- **SanDisk Dashboard.**
- **Samsung Magician.**
- **Kingston SSD Manager.**
- **Transcend RecoveRx.**

These are designed for verifying the health of authentic products.

Pros:

- Very accurate for genuine devices from the same manufacturer.
- Provide advanced metrics such as device temperature, SMART data, wear level, and life expectancy.
- Reliable, updated, and trusted sources.
- Useful for monitoring health of SSDs and branded storage devices.

Cons:

- Completely ineffective for generic, unbranded, or fake devices.
- Do not detect counterfeit products that mimic a brand but use different internal hardware.
- Mostly targeted at SSDs—not USB flash drives or memory cards.
- Still do not test full-capacity integrity or identify fake-reported storage sizes.
- Performance tests are limited to brand-specific metrics, not universal verification.

2.1.5. Community Tools and Fragmented Utilities

Apart from official and classic tools, users sometimes rely on multiple small utilities—speed testers, format tools, sector scanners—but none offer an integrated solution.

Pros:

- Some tools provide accurate niche functions (e.g., only speed testing).
- Open-source options increase transparency.
- Users can combine tools depending on their needs.

Cons:

- No single tool provides complete verification (capacity, integrity, performance) in one place.
- Users must switch between multiple programs, increasing complexity.
- Not all tools support all file systems or device types.
- Mixed quality, inconsistent results, and lack of standardized reporting.

2.2. Research Gap:

The evaluation of existing tools and storage device behavior reveals several important limitations that current systems fail to solve. These gaps highlight the need for a more reliable and comprehensive solution like CapScan.

2.2.1. Lack of Reliable Authenticity Verification Tools

Existing tools primarily test speed, storage integrity, or basic performance. However, none provide a strong method to verify a device's authenticity using hardware-level identifiers or manufacturer-confirmed signatures. This leaves users unable to confirm whether a device is genuine or altered [2].

2.2.2. No Access to Verified Hardware Identifiers

Most microSD cards and USB pendrives do not expose reliable identifiers such as CID, controller type, OEM ID, or true model information. During practical testing, many devices returned incomplete or inaccessible ID data, making authenticity verification through ID matching impossible [3].

2.2.3. Manufacturer Restrictions on Low-Level Information

Manufacturers often lock or hide controller metadata, firmware details, and internal structure to protect intellectual property. This prevents extraction of genuine identifiers that could help detect cloned or manipulated storage devices, creating an unavoidable verification gap [4].

2.2.4. No Standardized Database for Model & Controller Information

Unlike SSDs or HDDs, there is no global public database listing verified microSD or USB flash drive models, controller families, expected capacities, or manufacturer details. Without such a reference source, hardware authenticity cannot be cross-checked effectively [6].

2.2.5. Existing Tools Are Outdated or Limited on Large Capacity Devices

Legacy tools like H2testw, FakeFlashTest, F3, and MediaTester were designed for small capacities and often struggle with 128GB, 256GB, or larger modern drives. They are slow, unstable, or incapable of detecting advanced controller-level spoofing used in modern counterfeit devices [2].

2.2.6. Generic Operating System Tools Provide No Deep Validation

Built-in OS tools (Disk Management, Format Tool, Linux fdisk) only operate at the logical level. They cannot detect fake reported capacity, remapped sectors, manipulated firmware, or cloned NAND chips—leaving users with incomplete verification capabilities [5].

2.2.7. Inability to Retrieve Model ID from Small-Form Storage Devices

Unlike SSDs, microSD cards and USB pendrives do not expose model identifiers, production batches, or internal chip information. This makes it impossible to confirm whether the printed model number or branding matches the actual internal hardware, creating a significant authenticity gap [3].

2.2.8. Filesystem Conversion Limitations Reveal Hidden Defects

Many counterfeit or low-quality devices only function correctly with FAT32, while failing or corrupting data when converted to NTFS or exFAT due to weak controllers. Existing verification tools do not test filesystem-transition stability, leaving this issue undetected [5].

2.2.9. Lack of Unified Capacity and Authenticity Testing Tool

Current solutions either test capacity or provide limited device information, but no existing system combines capacity testing, metadata validation, filesystem behavior analysis, and anomaly detection in a unified workflow. This leaves a major gap in comprehensive counterfeit detection [6].

Chapter: 3 | SYSTEM ANALYSIS

3.1. Programming Language :

The project's primary programming language is Python. Because of its performance, flexibility, and support for object-oriented programming, Python was selected.

3.2. Environment for Integrated Development (IDE):

Visual Studio Code (VS Code): A popular, lightweight, open-source IDE used for writing and debugging Python programs. It provides features like syntax highlighting, IntelliSense, extensions, and an integrated terminal, making development efficient and user-friendly.

3.3. The Interpreter:

Python (Python): The project uses the standard Python interpreter to execute Python code. It supports a wide range of Python features, libraries, and optimizations, ensuring smooth and reliable program execution.

Hardware :

Custom Build – AMD Ryzen 5 5600, Samsung SSD 980 Pro 500GB SSD, 16GB RAM

This system, powered by a reliable Ryzen 5 5600 processor and paired with 16GB of dual-kit RAM and a fast Samsung 980 EVO Pro SSD which provides an excellent environment for Python programming & ensures smooth execution, testing, and compilation of code.

The hardware is more than capable of handling Python workloads & data processing.

Chapter: 4 | IMPLEMENTATION

4.1. Interface/Front-End Design

The front-end interface of CapScan is designed with a clean, dark-themed layout to ensure clarity, comfort, and ease of use. The application window is divided into two main sections: a left-side navigation panel and a central working area where device information, scan results, and system interactions occur.

Key Interface Components

- **Left Navigation Panel:**

A fixed sidebar offering quick access to core features:

- *Dashboard*
- *Devices*
- *Capacity Scan*
- *Device Health (SMART)*
- *History*
- *Exit*

These allow users to switch between functionalities seamlessly.

- **Dashboard Screen:**

Displays the software name *CapScan* along with a short tagline:

Verify capacity • Detect fake drives • Health checks

This acts as the home screen and provides a clear introduction.

- **Devices Screen:**

Shows a scrollable list of all connected storage devices, including:

- Drive letter
- Reported capacity
- Filesystem type
- Volume path

This panel allows users to quickly assess all detected devices in one view.

- **Action Buttons (Bottom Toolbar):**

Users can run key operations directly through the following buttons:

- **Refresh** – Reload connected device list
- **Quick Info** – Show essential metadata
- **Quick Capacity Scan** – Fast non-destructive capacity estimation

- **Full Capacity Validation (Destructive)** – Complete destructive write/read test
- **Central Display Area:**
A large console-style area that displays device lists, logs, and operation results in an easy-to-read text format.

The interface design focuses on simplicity, functional clarity, and fast user interaction.

4.2. Interface/Back-End Design

The back-end of CapScan interacts closely with the operating system to extract device information and perform capacity verification. The structure is designed for modularity and reliability.

Core Back-End Components

- **Device Enumeration Engine:**
Detects all attached storage devices and retrieves:
 - Drive letters
 - Total and available capacity
 - File system format (NTFS/FAT32/exFAT)
 - Volume information

This engine updates the Devices panel dynamically when the user clicks *Refresh*.

- **Quick Info Handler:**
Extracts non-destructive metadata such as:
 - Label
 - Capacity
 - Filesystem details
 - Allocation unit size

Results are displayed in the central console area.

- **Quick Capacity Scan Engine:**
Performs a lightweight estimation of true capacity without overwriting user data. Useful for fast checks but not as thorough as destructive validation.
- **Full Capacity Validation Engine (Destructive):**
Conducts a write-then-read verification of the entire storage space.
 - Detects fake capacity
 - Identifies data corruption
 - Reveals remapped blocks

This module requires wiping the device and is marked clearly in red for safety.

- **SMART Health Module (For Supported Devices):**
Gathers device health attributes, failure indicators, and controller-level logs. (Only available for devices that support SMART reporting.)

- **History Manager:**
Saves logs, previous scan summaries, and test outcomes for future reference.

4.3. Modules/Features

CapScan is organized into functional modules that directly align with the interface structure.

4.3.1. Dashboard Module

- Displays the application title and purpose
- Acts as an entry point to other features
- Helps users orient themselves in the application

4.3.2. Device Detection Module

- Lists all connected storage devices
- Shows drive letter, filesystem type, and capacity
- Supports refresh to update device list in real-time

4.3.3. Quick Info Module

- Provides high-level details about the selected device
- Non-destructive and safe for all storage types
- Helps users confirm system recognition of the device

4.3.4. Quick Capacity Scan Module

- Estimates the safe usable capacity
- Useful for detecting obvious fake devices quickly
- Significantly faster than full validation

4.3.5. Full Capacity Validation Module (Destructive)

- Performs comprehensive write/read tests
- Detects over-reported capacity
- Identifies corrupted sectors, unstable NAND, and suspicious patterns
- Suitable for identifying high-quality versus counterfeit devices

4.3.6. Device Health (SMART) Module

- Displays health metrics (when supported)
- Includes wear-level indicators, error counts, and lifespan predictions
- Helps determine if a device is degrading

4.3.7. History Module

- Stores previous test runs
- Allows users to track improvements or declines in device behavior
- Useful for documentation and record-keeping

4.3.8. Logging & Notification Module

- Tracks errors, warnings, and important system messages
- Displays logs directly in the central console area
- Helps users understand test outcomes clearly

Chapter: 5 | USER MANUAL

5.1. System Requirement

5.1.1. Hardware Requirement

Device - A Laptop/Desktop

Operating System - Microsoft Windows 7/10/11

RAM- minimum 4 GB

Storage-minimum 20 GB free space

5.1.2. Software Requirement

The required software for the project are :

IDE - Visual Studio Code (VS Code)

5.1.3. Required Pip/Library List in CapScan Project

- **tkinter** – GUI (Graphical User Interface) framework used to build the application interface
- **csv** – Used for reading and writing CSV files
- **datetime** – For handling date and time
- **hashlib** – For generating file/device hash values
- **math** – Mathematical operations
- **os** – File system and directory operations
- **platform** – For detecting system information
- **sqlite3** – Used for database operations
- **subprocess** – Running system commands
- **tempfile** – Creating temporary files
- **threading** – Managing background tasks
- **time** – Time-related functions (delays, timestamps)
- **Pillow** – Python imaging library used for opening, editing, and saving images in various formats.
- **psutil** – Cross-platform library that provides access to system and hardware information such as CPU, memory, disks, and processes.
- **PyQt5** – Comprehensive set of Python bindings used to create modern, interactive, and cross-platform graphical user interfaces (GUI).
- **pywin32** – Provides Python access to many Windows APIs, allowing interaction with system features such as files, registry, processes, and COM objects.

5.2. User Interfaces

Here is the detailed list and title of the project module with interface picture or screenshot.

Chapter: 6 | CONCLUSION

6.1. Conclusion

The CapScan project has successfully developed a reliable, user-friendly software tool designed to verify the authenticity, reliability, and performance of storage devices such as memory cards and USB drives. In a market increasingly affected by counterfeit and misreported storage products, CapScan addresses critical gaps by providing essential verification features—including true capacity testing, data integrity checks, and performance measurement. These capabilities allow users to identify deceptive or faulty storage devices before risking important personal, academic, or professional data.

Throughout the development process, CapScan focused on solving the limitations observed in existing tools, such as the inability to retrieve genuine device identifiers, outdated testing methods, unreliable results on modern high-capacity drives, and lack of unified verification approaches. By integrating multiple validation techniques into a single system, CapScan offers a more comprehensive and practical solution than traditional verification tools.

The final outcome is a dependable software application that enhances user confidence in storage device authenticity, reduces the risk of data corruption, and supports informed decision-making when handling digital storage media. CapScan establishes a foundation for future advancements, such as deeper hardware-level analysis, improved ID extraction methods, and expanded compatibility with modern storage technologies, ensuring long-term relevance and adaptability.

6.2. Limitations

Despite its effectiveness, the CapScan system has several limitations that influence the depth and accuracy of storage device verification:

- **Restricted Access to Hardware-Level Identifiers:**
Many microSD cards and USB drives do not expose essential identifiers such as CID, controller type, or OEM codes. As a result, CapScan cannot independently confirm the manufacturer or match devices against verified authenticity databases.
- **Limitations Imposed by Manufacturers:**
Several storage device manufacturers lock controller information or restrict access to low-level firmware data. This prevents CapScan from retrieving model-specific information and performing deeper authenticity validation.
- **Incomplete Support for Very Large-Capacity Devices:**
Certain high-capacity devices (256GB and above) may require extended testing durations, and some controllers behave unpredictably when performing full-scan verification.

- **Filesystem Conversion Constraints:**

Some counterfeit or unstable devices fail when converting from FAT32 to NTFS/exFAT. Although CapScan detects this behavior, it cannot override device-level controller defects or enforce filesystem stability.

- **No Centralized Database for Model Verification:**

The absence of a standardized global registry for microSD or USB flash drive models, controller families, and expected specifications limits the ability to cross-check authenticity with certainty.

- **OS-Level Restrictions:**

CapScan relies on operating system access policies. Some deeper read/write operations or device-level commands may be blocked depending on user privileges.

6.3. Future Works

To enhance the accuracy, functionality, and scope of CapScan, several improvements are planned for future versions:

- **Integration of a Device Authenticity Database:**

Development of a regularly updated online database containing verified model identifiers, controller types, expected capacities, and brand-specific authenticity markers.

- **Advanced Hardware-Level Identification Methods:**

Exploring advanced techniques such as direct controller interrogation, chip-off metadata extraction, and enhanced CID recovery for supported devices.

- **Improved Large-Capacity Device Support:**

Optimization of algorithms to reduce testing time and improve stability when scanning modern high-capacity flash drives (512GB, 1TB, etc.).

- **AI-Based Anomaly Detection:**

Implementing machine learning models capable of detecting suspicious behaviors—such as capacity spoofing, abnormal write patterns, or controller inconsistency—based on large datasets.

- **Cross-Platform Compatibility:**

Expanding CapScan beyond desktop environments to support Linux, macOS, and potentially mobile platforms for broader accessibility.

- **Deep Filesystem Stress Testing Tools:**

Adding automated stress tests that evaluate device stability across different filesystems (NTFS, exFAT, EXT4) to detect low-quality controllers or counterfeit NAND.

- **Enhanced User Interface and Reporting System:**

Future versions may include detailed PDF/HTML report generation, color-coded warning indicators, and more intuitive dashboards for non-technical users.

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Appendix A